Answer the following groups

Group 1 :

(A)Write the scientific term expressed by the following statement:

	Statement
1	It is the magnetic flux density which exerts a force of 1 N on a wire of 1m length carrying a current of intensity 1A placed perpendicularly to the magnetic field ……………………………………………………………………………………

(B) Put a sign (√) if the statement is correct and an (X) sign if the statement is incorrect:

	statement	
2	The force acting on a straight wire through which a current flows in a magnetic field decreases with increasing length of wire.	(……………)
3	The density of magnetic field produced by a solenoid carrying a current at a point on its interior axis increases when a soft iron bar inserted in it	(……………)
4	the magnetic torque is maximum value When the perpendicular to coil's plane normal to the magnetic field	(……………)
5	The magnitude of the dipole moment doesn't depend on the current intensity	(……………)

(C) Correct the underlined word

6 7	a) The rule used to determine the direction of the magnetic force is called <u>right hand screw rule</u> …………………………………………………………………………… b) The force acting on awire of length 1m carrying current of 20A placed in a magnetic field 0.1T is equal to <u>20N</u> ……………………………………………………………………………

(D) Put circle around the correct answer

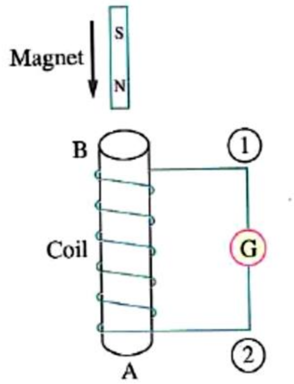
8	**A solenoid carrying current has a magnetic flux density (B) at a point on its axis. If it is stretched to double its length with same current intensity, then magnetic flux density at a point on its axis is:** **A) B** **B) ½B** **C) 2B** **D) 4B**
9	**The magnetic flux density at a point located vertically at a distance d from a straight wire carrying a current of intensity I equals B. When increasing the current intensity to double, then the magnetic flux density becomes...** **A) 0.5B.** **B) 2B.** **C) B.** **D) 4B.**
10	**What is the magnetic field at the center of a circular loop of wire of radius 4cm when a current of 2A flows in the wire? (where : $\mu = 4\pi \times 10^{-7}$ Wb/A.m)** **A) $\pi \times 10^{-7}$ T** **B) $\pi \times 10^{-5}$ T** **C) $\pi \times 10^{-4}$ T** **D) $\pi \times 10^{-3}$ T**

Group 2 :

A) Put circle around the correct answer

11	When the number of turns in the dynamo's coil is doubled, then the maximum generated electromotive force generated from it: a)Increases to double b) Decreases to its half value c) Remained constant d) Increases 4 times
12	A bar magnet is dropped into a solenoid connected to a galvanometer, as shown in the opposite figure. Which choice of the following is correct?

	The magnetic pole formed at A	The magnetic pole formed at B
a)	North	North
b)	South	South
c)	South	North
d)	North	South



B) Mention the mathematical formula for each of the following

13	Mutual induction coefficient	…………………………………..
14	average induced emf in a coil	………………………………………

C) Compare between the following:

15 16	Fleming's right hand rule	Lenz rule
Uses	…………………………………..	………………………………………

D) Write the scientific term expressed by the following statements:

17	**induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect** ………………………………………………………………………………… ……………………………………………………………………………………
18	**The phenomenon of inducing emf in a coil due to change in current in the same coil and hence the change in magnetic flux in the coil** ………………………………………………………………………………… ……………………………………………………………………………………

(E) Put a sign (√) if the statement is correct and an (X) sign if the statement is incorrect:

	statement	
19	**The ratio of output electric power to the input electric power in the electric transformer is called efficiency of transformer**	(……………)
20	**If the wires moves by an angle along the direction of the magnetic field an induced emf generated between two ends of the wire is given by : $\text{emf} = \frac{B\,l\,v}{\sin\theta}$**	(……………)

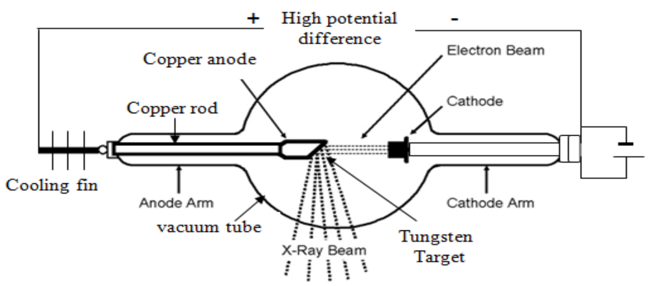
Group 3 :

A) Correct the underlined word in each of the following

	Statement
21	The wavelength which is associating the maximum intensity of the radiation (λ_m) is <u>directly</u> proportional to the temperature of a radioactive source on Kelvin scale. ………………………………………………………………………………
22	Wavelengths of invisible electromagnetic rays lie between the wavelengths of ultraviolet rays and gamma rays and is characterized by a short wavelength is called <u>infrared</u> rays ………………………………………………………………………………
23	metastable energy level characterized by its lifetime is relatively long nearly <u>10^{-8} s</u> ………………………………………………………………………………

B) In the opposite figure: Mention the name of the device and mention one use.

24 25	……………………………………… ………………………………………



C) Put circle around the correct answer

26	The energy of the cathode rays equals ………… a. $h\upsilon$ b. $\frac{1}{2}mv^2$ c. mv d. 2 mv
27	The ratio between the frequencies of photons is 1 : 2. So, the ratio between their energies is a. 1 : 1 b. 2 : 1 c. 1 : 2 d. 4 : 1
28	When a photon collides with a free electron so, ………….. a. the photon loses its energy. b. the electron acquires kinetic energy equals the photon energy. c. the frequency of the scattered photon is less than the frequency of the incident photon. d. the photon and the electron move together.

(D) Put a sign (√) if the statement is correct and an (X) sign if the statement is incorrect:

	statement	
29	**To free (emit) an electron with kinetic energy from the metal, the energy of the incident photon must be less than the work function**	(……………)
30	**The Scientific idea of cathode ray tube is depend on the thermionic emission**	(……………)

Group 4 :

A) Correct the underlined word in each of the following

31	**A semiconductor crystal is doped with impurities from pentavalent element so the concentration of free electrons (n) is <u>smaller than</u> the concentration of holes (p).** ………………………………………………………………………………………
32	**<u>Digital electronics</u> deal with natural quantities and convert them to continuous electrical signals).** ……………………………………………………………………………
33	**in transistor the ratio between the collector current and the base current is <u>less then</u> one** …………………………………………………………………………
34	**in thermal dynamic equilibrium the number of bonds broken per second will be <u>greater</u> the number of bonds mended per second.**

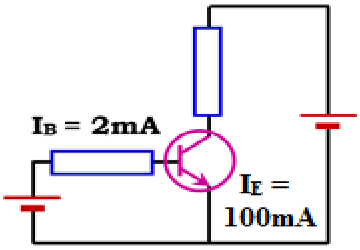
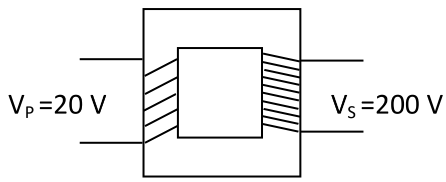
Write the name of the logic gate in each of the following cases

35	**Logic gate has one input.** ……………………………………………………………………………………
36	**Logic gate has two inputs and gives high output if the voltage of one of them is high and the other is low.** ……………………………………………………………………………………

Compare between each of the following

<table>
<tr><td rowspan="2">37
38</td><td>Point of comparison</td><td>NOR gate</td><td>NAND gate</td></tr>
<tr><td>Complete the truth table</td><td><table><tr><td>Input A</td><td>Input B</td><td>Output</td></tr><tr><td>1</td><td>1</td><td>……</td></tr></table></td><td><table><tr><td>Input A</td><td>Input B</td><td>Output</td></tr><tr><td>1</td><td>1</td><td>…….</td></tr></table></td></tr>
<tr><td rowspan="2">39
40</td><td>Point of comparison</td><td>In the forward connection</td><td>In the reverse connection of P-N junction</td></tr>
<tr><td>Electric Resistance of P-N junction</td><td></td><td></td></tr>
</table>

Group 5: Answer the following problems

41 42	**Mention two** الصفحة رقم (6) انتهت الاسئلة
43 44	**If the critical frequency of zinc is $\frac{3\times10^{8}}{3000\times10^{-10}}$ Hz. Calculate its work function.(Planck's constant = 6.625×10^{-34} J.s)**
45 46	**A rectangular loop of length 0.08 m, width 0.05 m and 100 turns is placed in a magnetic field of 10 Tesla flux density, carrying current 1 A, Find The maximum torque acting on the coil**
47 48	**From the opposite figure:Findthe value of I_C** 
49 50	**Is this step-up or step-down transformer ?** 

(انتهت الاسئلة)